BeeQ Response to Reviewers and Summary of Revisions

Manuscript: BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

We thank the reviewers for their careful assessment and constructive suggestions. The revision gives greater prominence to toxic-compound detection, adds a quantitative comparison of molecular representations, expands the MLP reporting, and specifies directions for kernel adaptation. The responses below describe the changes and identify comments that remain partially addressed.

Page references refer to the eight-page revised manuscript, BeeQ_FinalV01.pdf. Related weaknesses and recommendations are addressed together; Reviewer 1 follows the five recommended improvements, and Reviewer 2 follows the two recommended improvements and the third weakness.

Reviewer 1

Comment 1 Emphasis on toxic compound sensitivity

Comment. *Give the sensitivity drop on new data equal billing with the quantum-vs-classical comparison in the abstract and conclusion, it may be the more useful takeaway for readers who want to actually use this.*

Response. We revised the closing synthesis of the Abstract and the opening of the Conclusions to state two main findings explicitly: the evaluated quantum kernels did not provide a consistent improvement over matched RBF, and sensitivity for toxic compounds decreased outside Development. The Introduction now identifies toxic-compound detection as an explicit purpose of the transfer assessment. Previously, the sensitivity decrease was reported, but the final synthesis emphasized kernel geometry.

Section III-B now adds the pooled out-of-fold sensitivity on Development, 0.338-0.410, alongside the historical-holdout range of 0.082-0.265, and reports that the models missed 36-45 of the 49 toxic compounds. This makes the practical detection limitation more explicit.

Location. Abstract, p. 1; Section I, final paragraph, p. 2; Section III-B, p. 5; Section IV, opening paragraph, p. 6.

Comment 2 Connection to kernel concentration

Comment. *Add a sentence connecting the results to the known "kernels flatten out" theory directly, rather than leaving readers to make that connection themselves.*

Response. The Introduction already cited Thanasilp et al. [4] on exponential concentration, and this discussion is retained. Section III-D also retains the analysis of geometric alignment, the small-angle Gaussian approximation, and the bandwidth-matched effective-rank comparison. These findings support a limited distinction between the evaluated kernels under the present representation and parameter settings; they do not establish exponential concentration as the cause.

The revision does not add the requested explicit connection in the Discussion. This aspect of the comment therefore remains only partially addressed, and we do not claim to have demonstrated concentration or its scaling with qubit count.

Location. Existing discussion in Section I, p. 1, and Section III-D, pp. 5-6; reference [4], p. 8. No new concentration-specific discussion was added.

Comment 3 Previous inspection of the historical holdout

Comment. *Flag the historical holdout's earlier exposure as a real limitation in the same breath as praising the provenance of the Costa Rica panel, so readers know to trust the two holdouts differently.*

Response. The manuscript continues to identify the historical holdout as a set inspected during earlier exploratory stages. Section II-C retains that disclosure, and the final paragraph of the Introduction continues to distinguish this set from the eight-compound external challenge. These disclosures were already present in the previous version.

The revised manuscript does not add an account of what was inspected or whether that exposure could have influenced descriptor selection, curation, or other methodological decisions. The request for a fuller limitation statement is therefore not fully addressed. Prior inspection limits the interpretation of the historical holdout as an untouched external test, even though the reported final fitting and threshold-selection procedure uses Development data.

Location. Section I, final paragraph, p. 2; Section II-C, p. 2; Section II-D, p. 3, for the retained fitting and threshold-selection procedure.

Comment 4 Methods for adapting quantum kernels

Comment. *Mention existing methods for fixing or adjusting quantum kernels so they don't flatten out, since the conclusion already points in that direction without citing anything specific.*

Response. We replaced the broad future-work paragraph with concrete directions for kernel adaptation: extending encoding-scale optimization, training embedding parameters through kernel-target alignment, and evaluating projected kernels based on local quantum observables. The revised paragraph proposes initially keeping X10 fixed to distinguish changes to the kernel from changes to the representation, and evaluating these approaches under nested molecular-cluster validation with attention to toxic-compound recovery and false positives.

We also added Shaydulin and Wild, "Importance of kernel bandwidth in quantum machine learning," as reference [28]. These additions describe future experiments; the present revision does not report an implementation or evaluation of these adaptive or projected kernels.

Location. Section IV, final paragraph, p. 7; new reference [28], p. 8.

Comment 5 MLP class balancing and interpretation

Comment. *Rerun the MLP model with the same class balancing used for the others or clearly flag the mismatch as a limitation before attributing its worse results to the model itself.*

Response. We expanded the MLP description with StandardScaler, ReLU, Adam, max_iter = 2000, native early stopping, a 12-candidate search, and the five-outer/four-inner molecular-cluster-disjoint protocol. Table I changes its AUROC from 0.677 ± 0.125 to 0.679 ± 0.100, AUPRC from 0.596 ± 0.138 to 0.594 ± 0.112, balanced accuracy from 0.654 to 0.632, and MCC from 0.428 to 0.338.

Section III-B adds a holdout AUPRC of 0.329, with 13 of 49 toxic compounds identified and 22 false positives among 132 non-toxic compounds. However, class or sample weighting remains unspecified, and the earlier unweighted-training disclosure has been removed. A matched class-balancing experiment therefore cannot be confirmed from this account. Section III-B also retains older MLP values that require reconciliation.

Location. Section II-D, p. 3; Table I, p. 4; Section III-B, p. 5.

Reviewer 2

Comment 1 Quantitative context for X10

Comment. *Benchmark X10 against prior ApisTox results. Include a direct quantitative comparison (even a summary table) of BeeQ's AUROC/AUPRC against previously published fingerprint-, graph kernel-, and GNN-based models on ApisTox, so readers can judge whether the compact 10-descriptor representation is competitive or represents a deliberate trade-off in interpretability/qubit-compatibility versus raw performance.*

Response. We added Section III-F and Table II to provide quantitative context for X10. The table reports AUROC for WL-OA with SVM (0.805 ± 0.073), ECFP with Random Forest (0.787 ± 0.091), RDKit2D descriptors with Random Forest (0.784 ± 0.064), Mordred with Random Forest (0.755 ± 0.103), atom counts with Random Forest (0.748 ± 0.068), and MACCS with Random Forest (0.732 ± 0.104), alongside X10 with Random Forest (0.748 ± 0.057). The previous version cited the ApisTox benchmark but contained no corresponding quantitative comparison.

The new text describes X10 as a deliberately compact representation and explains its one-descriptor-per-qubit design. Section III-F presents the added comparisons as evaluations under the same validation procedure and cites [9]. Table II does not separately identify published versus newly evaluated results, and it does not include AUPRC or a GNN result. It therefore adds useful quantitative context but does not cover every part of the requested published-benchmark comparison.

Location. New Section III-F, p. 6; new Table II, p. 7; retained reference [9], p. 8.

Comment 2 Representations beyond X10

Comment. *Test representation dimensionality beyond X10. Run the same pipeline with an expanded descriptor set (e.g., X15, X20, or a full fingerprint baseline) to disentangle whether the observed sensitivity limitations and quantum-classical equivalence stem from the learning method or from the compact representation forced by the 10-qubit constraint.*

Response. The new comparison extends the reported classical baselines beyond X10 to fingerprints, larger descriptor families, atom counts, and a graph kernel. In particular, ECFP, RDKit2D, and Mordred provide alternatives to the compact descriptor set, while the accompanying discussion explains why the quantum comparison retains ten descriptors and ten qubits.

The revision does not report an X15 or X20 experiment, an expanded-representation quantum-kernel comparison, or corresponding sensitivity and transfer results for the added representations. The new AUROC comparison therefore broadens the representation context, but it does not establish whether representation dimensionality causes the sensitivity decrease or the similar quantum and classical performance. Descriptor selection and further controlled comparisons remain future work.

Location. New Section III-F, p. 6; new Table II, p. 7; Section IV, final paragraph, p. 7.

Comment 3 Scope of conclusions from CR8

Comment. *CR8 conclusions rest on n=2 toxic compounds. While the authors appropriately caveat this in the Results, the Conclusion section restates the CR8 failure with a rhetorical weight that outpaces what two data points can statistically support, even though it remains a valid qualitative observation.*

Response. We strengthened the limitation on interpretation of CR8. Section III-E now explicitly states that the two toxic compounds support compound-level observations rather than population-level conclusions. The Conclusions also state that eight compounds, including only two toxic compounds, do not support an estimate of population sensitivity. This extends the small-sample qualification that was already present in the earlier Results.

Section III-E additionally reports that the revised MLP correctly identified Isocycloseram and produced false positives for Fluoxapiprolin and Florylpicoxamid. However, the Abstract, Conclusions, and Figure 4 retain the earlier account. The small-sample qualification has been strengthened, but those conflicting descriptions need to be reconciled before the revised CR8 outcomes can be summarized consistently.

Location. Section III-E, p. 6; Section IV, p. 7. Conflicting earlier descriptions remain in the Abstract, p. 1, and Figure 4 and Section IV, p. 7.

Additional editorial changes

The revised manuscript replaces the anonymous author placeholders with names, affiliations, and email addresses, and adds the conference heading. Equations (2)-(4) are now numbered, and the unresolved reference in the geometric discussion has been replaced by a reference to Equation (4). These are editorial changes in addition to the substantive revisions described above.